

For my EDA poster, I focused on identifying the visual correlations between digital habits and academic success. I discovered that students with higher 'Short-Form Video' usage consistently had lower study hours and productivity scores.

To take this analysis further, I created a multiple linear regression model. My goal is to move beyond just looking at the data, but to build a framework that can predict a student's productivity score based on short-form video hours, social media hours, and study hours.

Response Variable (Y): Productivity Score

Predictory Variables (X):

- X1: short form video hours
- X2: social media hours
- X3: study hours

Hypothesis Testing:

- Null Hypothesis (H_0): Short-form video usage has no effect on a student's productivity score ($\beta_1=0$)
- Alternative Hypothesis (H_a): Short-form video usage significantly decreases a student's productivity score ($\beta_1<0$)
- We are looking for a P-value <0.05 to reject the null hypothesis

Predictor Variable	Coefficient(Weight)	Standard Error	P-Value
Intercept(Base Score)	8.006392	0.00832	0.000
Short Video Hours	-0.2464	0.0047	0.000
Social Media Hours	-0.2536	0.0024	0.000
Study Hours	0.08906	0.00028	0.000

Productivity Score= $8 - 0.2464(X_1) - 0.2536(X_2) + 0.08906(X_3)$

Our model shows that Social Media Hours (-0.2536) and Short-Form Video (-0.2464) act as a dual-threat to productivity. However, I have focused my narrative on Short-Form Video because it correlates more strongly with reduced Attention Spans and the Brain Rot Index. For every additional hour spent watching short videos/social media, a student's productivity score is expected to drop by -0.2464 and -0.2536 points, holding all other variables constant. Since the P value is $0 < 0.05$ - we reject the null hypothesis.

3	<i>Regression Statistics</i>	
4	Multiple R	0.56802885
5	R Square	0.32265677
6	Adjusted R Square	0.32265271
7	Standard Error	1.0507398
	Observation	499999

-The Adjusted R-Square of 0.322 means that our model explains 32.2% of the variance in student productivity. While other factors like sleep and environment matter, digital habits and study time account for nearly a third of a student's academic success.

Standard Error: 1.05

- This indicates that when this model is utilized to predict a student’s score, the prediction is typically off by about 1.05 points.
- Therefore, this model is a “good fit”, as it does acknowledge that while digital habits are a huge factor, it is not the only factor. This leaves room for other things like sleep or mental health.